

# ONTARIO TECH UNIVERSITY

Faculty of Business and Information Technology  
MBAI 5600G - Applied Integrative Analytics Capstone Project

## MILESTONE 2

### Literature Review & Data Source Identification

S&P 500 Stock Prediction Using Deep Learning and Sentiment Analysis

**Group Number:** Group 6

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## 1. Updated Executive Summary

This project addresses one of the most challenging problems in applied machine learning: predicting stock price movements in the US equity market. Specifically, the goal is to build a dual-output predictive system for all S&P 500 constituent stocks that simultaneously forecasts the next-day closing price (a regression task) and generates a binary buy or sell trading signal (a classification task). The system aims to support actionable, data-driven investment decisions for retail investors and quantitative analysts.

Our literature review examined 20 scholarly articles spanning 2001 to 2026, with a strong focus on research published between 2024 and 2026. The review covered five key themes: deep learning for financial time series, sentiment analysis and NLP in finance, hybrid and ensemble modelling approaches, feature engineering and technical indicators, and model explainability and financial performance validation. The review confirmed that combining technical indicators with NLP-based sentiment features consistently improves predictive accuracy, that hybrid recurrent architectures outperform standalone models, and that Transformer-based models represent the current frontier in sequence modelling. However, most existing studies focus on a single stock or small stock universes, rarely validate predictions using real financial metrics, and offer limited model transparency.

This project bridges those gaps by applying five models - LSTM+GRU, BiLSTM, CNN+LSTM, Time Series Transformer, and Random Forest - to all S&P 500 stocks simultaneously, integrating FinBERT-based sentiment features from Yahoo Finance News, and validating performance using both standard ML metrics and financial indicators including Sharpe Ratio, P&L simulation, and backtesting. All data is sourced from Yahoo Finance via the yfinance Python library, which is free, reliable, and requires no API key. Backup datasets from Alpha Vantage, Kaggle and Quadl are identified for resilience. The project is feasible within the available timeline and hardware using Google Colab and Kaggle Notebooks for GPU - accelerated training.

## 2. Literature Review

This section reviews the existing research most relevant to our project. Rather than summarizing each article individually, we organize the review into five key themes: (1) deep learning for stock price prediction, (2) sentiment analysis and NLP in finance, (3) hybrid and ensemble modelling approaches, (4) feature engineering and technical indicators, and (5) model explainability and financial performance validation. This structure allows us to identify trends, compare methodologies, and clearly position our project's contributions.

### 2.1 Deep Learning for Stock Price Prediction

Long Short-Term Memory (LSTM) networks have become the dominant approach for stock price prediction over the past decade, primarily because they are designed to learn from sequential data and remember long-term dependencies - both critical properties for financial time series. Early foundational work by Fischer and Krauss [1] demonstrated that LSTM models significantly outperform classical statistical methods including Random Walk, deep neural networks, and logistic regression on S&P 500 constituent stocks from 1992 to 2015, achieving daily returns of 0.46% with a Sharpe ratio of 5.8. This established LSTM as the primary baseline recurrent architecture for equity prediction tasks.

Selvin et al. [2] compared RNN, LSTM, and CNN architectures on Indian NSE stock data using a sliding window approach, finding that all three deep learning models significantly outperformed ARIMA. Importantly, CNN demonstrated superior ability to capture sudden dynamic changes in stock market behaviour due to its focus on the current input window rather than historical dependencies - a finding that directly motivates our inclusion of CNN as a component in our CNN+LSTM hybrid model.

More recent research has moved beyond plain LSTM toward hybrid recurrent architectures. Farhadi et al. [3], our base research article, combined LSTM and GRU layers into a single sequential model targeting the Tehran Stock Exchange, demonstrating measurable improvements in MSE over standalone LSTM or GRU using 42 technical indicators across four categories. Salokhe et al. [4] further validated this direction in 2024, comparing LSTM and GRU architectures on historical stock data from Yahoo Finance and confirming that both models effectively capture temporal dependencies and complex patterns in financial time series, with GRU offering a simpler design and fewer parameters while LSTM excels at longer-range patterns.

Bidirectional LSTM (BiLSTM) has gained significant traction as a natural extension of standard LSTM. Qiu et al. [5] demonstrated that an attention-based LSTM model using wavelet denoising on S&P 500 and DJIA datasets achieved a coefficient of determination above 0.94 and MSE below 0.05, outperforming standalone LSTM, LSTM with denoising, and GRU - validating the value of attention mechanisms and richer sequential context in financial prediction. Liu et al. [21] further extended this direction with DSTP-RNN, a dual-stage two-phase attention mechanism applied to multivariate time series, demonstrating that separating spatial and temporal attention stages yields measurably lower prediction error than single-stage attention models - supporting the value of structured attention in our Transformer and BiLSTM designs. Kim and Won [6] proposed a hybrid LSTM model combined with multiple GARCH-type models for stock price volatility forecasting on the KOSPI 200 index, demonstrating that the GEW-LSTM hybrid achieves 37.2% lower MAE than the best existing hybrid model - validating the effectiveness of combining LSTM with complementary statistical models for financial time series.

The Transformer architecture has recently been adapted for financial time series with strong results. Qian [7] proposed an incremental learning-based enhanced Transformer specifically for online stock price prediction, using multi-head self-attention and continual normalization to handle non-stationary financial data streams across five publicly available datasets. Chen [8] compared five distinct Transformer structural variants on the S&P 500 index using data from May 2015 to May 2024, finding that the decoder-only Transformer outperforms

all other variants across 5, 10, and 15-day sliding windows - directly informing our window design and Transformer architecture choice.

A key limitation across most of these studies is their narrow scope: the majority focus on one or a small number of stocks, and very few validate their models using actual financial performance metrics. This is a gap our project directly addresses by targeting all S&P 500 stocks and evaluating models using Sharpe Ratio, P&L simulation, and walk-forward backtesting.

## 2.2 Sentiment Analysis and NLP in Financial Prediction

A growing body of research demonstrates that market sentiment - derived from news articles, earnings calls, and social media - carries significant predictive power beyond what price and volume data alone can provide. Araci [9] introduced FinBERT, a BERT-based model fine-tuned on financial text, which has since become the standard benchmark for financial sentiment classification, achieving over 86% accuracy on financial news sentiment tasks and establishing the foundation for the sentiment pipeline in our project.

Das et al. [10] conducted a comprehensive survey on integrating sentiment analysis with graph neural networks for stock prediction and found that hybrid architectures combining price features and sentiment scores consistently outperform price-only models by 3 - 8% in directional accuracy. Li et al. [22] demonstrated this effect empirically on Hong Kong stock market data, showing that models incorporating both stock prices and news sentiment significantly outperform price-only baselines, and that sentiment impact varies by company size and news volume - a finding that directly informs our decision to flag low-news days with a binary indicator feature. Their survey identifies that the combination of structured price signals and unstructured news-derived sentiment is one of the most productive directions in current financial ML research - directly motivating our feature engineering approach.

Gu et al. [11] demonstrated that a FinBERT-LSTM architecture applied to NASDAQ-100 stocks using Benzinga news articles categorized by market, industry, and stock level consistently outperformed standalone LSTM and deep neural network baselines, evaluated using MAE, MAPE, and Accuracy. Their finding that stock-level news contributes more predictive power than market-level news directly informed our decision to extract per-ticker Yahoo Finance News headlines rather than relying on market-wide news feeds.

Karadas et al. [12] proposed a multimodal stock prediction framework that fuses financial metrics, tweets, and news articles using FinBERT for sentiment analysis with an LSTM base model, demonstrating up to 5% improvement in forecast effectiveness over the reference model on three S&P 500 stocks. Our project builds on this direction by additionally creating interaction features between sentiment scores and price momentum signals - a feature engineering strategy not fully explored in prior work. One notable gap in the sentiment literature is the limited use of news-derived sentiment at large stock universe scale; most studies test sentiment models on 5 - 30 stocks, and our project addresses this directly by applying FinBERT across all S&P 500 tickers.

## 2.3 Hybrid and Ensemble Modelling Approaches

The use of ensemble and hybrid models has become increasingly common in financial prediction research. Khaidem et al. [13] demonstrated that Random Forest using technical indicators including RSI and stochastic oscillators consistently achieves strong directional accuracy on equity prediction tasks, outperforming existing forecasting algorithms with encouraging out-of-bag error estimates. This establishes Random Forest as the standard interpretable benchmark against which deep learning models should be compared.

Qian [7] showed that Transformer-based models with incremental learning handle non-stationary financial data effectively, while Chen [8] confirmed through systematic comparison that structural choices in Transformer design significantly impact stock prediction performance. Wei et al. [14] proposed CEEMD-CNN-BiLSTM-AM, a hybrid model that applies Complete Ensemble Empirical Mode Decomposition before CNN-BiLSTM with an attention mechanism, demonstrating strong performance on Chinese stock data and particularly excelling at identifying low-frequency price trends in bull market conditions - directly supporting our inclusion of BiLSTM and CNN+LSTM as Models 2 and 3.

A recurring limitation in the hybrid model literature is the lack of systematic model comparison under identical experimental conditions. Nabipour et al. [23] conducted a systematic comparison of RNN, LSTM, and several ensemble methods on stock market prediction tasks, finding that deep learning models consistently outperform traditional machine learning approaches when sufficient historical data is available, while also highlighting that no single architecture dominates across all market conditions - directly motivating our five-model comparative framework. Studies typically propose one or two architectures and compare them against simple baselines, rather than conducting a rigorous side-by-side evaluation of multiple advanced models. Our project explicitly addresses this by training all five models on identical feature sets and evaluating them under the same train-validate-test split and evaluation metrics, enabling a fair and reproducible comparison.

## 2.4 Feature Engineering and Technical Indicators

Feature engineering is consistently identified as one of the most impactful steps in financial ML pipelines. A rich set of technical indicators - including RSI, MACD, Bollinger Bands, ATR, and OBV - carry significant predictive signal when used as model inputs. Farhadi et al. [3] demonstrated that models trained on 42 technical indicators across four categories (trend, momentum, volatility, volume) substantially outperform models trained on raw OHLCV data alone, directly motivating the comprehensive indicator set in our feature engineering pipeline.

Lag features and rolling statistical features are well-established complements to technical indicators. Bao et al. [15] proposed a deep learning framework combining wavelet transforms, stacked autoencoders, and LSTM for stock price forecasting on six market indices, demonstrating that hierarchically extracting deep features from denoised time series - including price lags and structural decomposition - significantly outperforms models that rely on raw closing prices alone. Sun et al. [16] proposed the StockMixer with ATFNet model integrating both time-domain and frequency-domain features, demonstrating that multivariate inputs combining price-based, volume-based, and spectral features consistently improve prediction accuracy and Sharpe Ratio in backtesting - providing strong justification for our 60+ feature engineering pipeline.

A more novel contribution in recent literature is the use of interaction features - products of two or more signals. Chen [8] found that combining structural attention signals with momentum features provided additional predictive power beyond individual components. Karadas et al. [12] created interaction features between sentiment scores and price data, finding that combined signals outperform individual contributions. Our project expands this concept by creating four distinct interaction features: sentiment  $\times$  price momentum, sentiment  $\times$  volume, net sentiment  $\times$  RSI, and net sentiment  $\times$  MACD - a feature engineering contribution not fully explored in any prior study reviewed.

## 2.5 Model Explainability and Financial Performance Validation

As AI-driven trading systems become more widely used, model interpretability has become a critical requirement. Lundberg and Lee [17] introduced SHAP (SHapley Additive exPlanations), which provides consistent, theoretically grounded feature importance scores for any ML model. Muhammad et al. [18] applied both SHAP and LIME across four diverse stock market datasets in 2024, demonstrating that a deep learning model achieves 94.9% average accuracy and 94.85% F1-score, and that SHAP-selected top 10 features maintain performance

while reducing model complexity - directly motivating our use of SHAP for Random Forest feature importance analysis and feature selection.

On the Transformer side, Vig [19] showed that attention weights in transformer-based models serve as a meaningful form of interpretability, allowing users to see which time steps the model focuses on when making predictions. The open-source visualization tool introduced covers multiple attention scales including model view, head view, and neuron view, providing qualitative insights about model behaviour on sequential data that complement SHAP-based quantitative analysis.

Financial performance validation is a critical gap in most academic studies. Sezer et al. [20] conducted a comprehensive survey of deep learning methods for financial time series from 2005 to 2019 and found that fewer than 20% of papers validated their models using any form of financial metric beyond classification accuracy or regression error. Sun et al. [16] is a notable recent exception, reporting Sharpe Ratio and backtesting IC metrics alongside standard ML metrics. Our project addresses this gap systematically by evaluating all five models using Sharpe Ratio, total P&L, maximum drawdown, and win rate through walk-forward backtesting on the held-out test set - directly responding to the validation gap identified by Sezer et al. [20].

2.6 Research Gap Analysis

Based on the literature review above, we identified the following key gaps that our project directly addresses:

| Gap Identified                | How It Appears in Literature                                                  | How Our Project Addresses It                                                              |
|-------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Narrow stock universe         | Most studies test on 1 - 30 stocks; findings may not generalize [3][5][12]    | We target all ~500 S&P 500 constituent stocks                                             |
| No financial validation       | < 20% of studies report Sharpe Ratio or P&L metrics [20]                      | Sharpe Ratio, P&L simulation, and backtesting are core evaluation components              |
| Single model focus            | Most papers propose one architecture vs. simple baselines only [3][6][14]     | We rigorously compare 5 architectures under identical conditions                          |
| Limited sentiment integration | Sentiment tested on small stock universes; no interaction features [10][11]   | FinBERT sentiment + 4 novel interaction features across all S&P 500 tickers               |
| Weak explainability           | Few financial ML papers include feature importance or attention analysis [20] | SHAP values for Random Forest + attention weights for the Transformer                     |
| Dual output models are rare   | Most papers predict price or direction, not both simultaneously               | Our system produces both next-day price (regression) and buy/sell signal (classification) |

### 3. Data Source Identification

This section identifies and evaluates the datasets we will use for this project. All data sources have been selected based on reliability, accessibility, coverage, and compatibility with our modelling objectives.

#### 3.1 Primary Dataset

Our primary data source is Yahoo Finance, accessed through the yfinance Python library. This source provides all the price and news data required for both regression and classification tasks.

| Attribute        | Details                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------|
| Source           | Yahoo Finance via yfinance Python library (v0.2.x)                                                      |
| Price Data       | Historical daily OHLCV: Open, High, Low, Close, Adjusted Close, Volume for all S&P 500 stocks           |
| News Data        | Yahoo Finance News headlines per ticker - used for FinBERT sentiment scoring                            |
| Time Range       | January 2015 - December 2024 (approx. 10 years, ~2,500 trading days per ticker)                         |
| Universe         | All S&P 500 constituent stocks (~500 tickers). Tickers with fewer than 3 years of data will be excluded |
| Volume           | ~500 tickers × 2,500 days = ~1.25 million rows of price data; thousands of news articles per ticker     |
| Access Method    | yfinance.download() for price data; yfinance.Ticker(symbol).news for news headlines                     |
| Data Format      | Pandas DataFrame (in-memory); exportable to CSV for persistence                                         |
| Cost / License   | Free - no API key required; data provided for personal and research use                                 |
| Target Variables | Regression: next-day closing price. Classification: buy (1) if next-day return > 0, else sell (0)       |

#### 3.2 Backup Datasets

| Source                         | Data Provided                                                                         | When to Use                                                                    |
|--------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Alpha Vantage API              | OHLCV + structured news sentiment endpoint; free tier: 500 API calls/day              | Primary backup if yfinance rate limits become restrictive during bulk download |
| Kaggle: S&P 500 Stocks Dataset | Pre-downloaded OHLCV for all S&P 500 stocks; no internet dependency                   | Offline backup if API access fails entirely; useful for rapid prototyping      |
| Quandl / Nasdaq Data Link      | Historical US equity prices and adjusted close; free academic tier available          | Secondary price data backup; well-maintained and clean historical data         |
| NewsAPI.org                    | Financial news headlines from Reuters, Bloomberg, CNBC; 100 requests/day on free tier | Backup sentiment source if Yahoo Finance News coverage is insufficient         |

3.3 Dataset Description and Structure

| Attribute                   | Details                                                                              |
|-----------------------------|--------------------------------------------------------------------------------------|
| Number of Rows              | ~1.25 million rows of price data (500 tickers × ~2,500 daily observations each)      |
| Number of Features (Raw)    | 6 base columns: Date, Open, High, Low, Close, Volume (per ticker)                    |
| Engineered Features (Total) | ~60+ features after full feature engineering pipeline                                |
| Target: Regression          | Next-day Adjusted Close price (continuous variable)                                  |
| Target: Classification      | Binary buy/sell signal: 1 if next-day return > 0%, else 0                            |
| News Articles               | Estimated 5,000 - 50,000+ articles per ticker over 10 years (varies by company size) |
| Data Format                 | CSV files per ticker for price data; JSON for news headlines (via yfinance API)      |
| Missing Data                | Non-trading days (weekends, holidays) are absent - standard for daily equity data    |

## 4. Preliminary Data Assessment

Before full preprocessing begins in Milestone 3, we have conducted a preliminary evaluation of the dataset to assess its quality, completeness, and suitability for our modelling objectives.

### 4.1 Data Completeness

Yahoo Finance via yfinance provides complete daily OHLCV data for all current S&P 500 constituent stocks since at least 2015. Some stocks joined the index after 2015 and will have shorter histories. Any ticker with fewer than three years of data will be excluded, which is expected to affect fewer than 5% of tickers.

### 4.2 Missing Value Analysis

- Non-trading days (weekends and public holidays): Simply absent from the dataset - not missing values - handled by using only trading-day observations in all feature computations.
- Adjusted close data: A small number of tickers may have inconsistencies around stock splits, corrected using yfinance's built-in adjustment flags.
- News data sparsity: Smaller-cap companies may have no news on some trading days. These will be set to neutral sentiment (0.5) and flagged with a binary 'no-news' indicator feature.
- Corporate actions: Dividend payments, mergers, and delistings can introduce anomalies. Adjusted close prices from yfinance account for dividends and splits, reducing this risk.

### 4.3 Feature Relevance and Class Distribution

The six raw OHLCV features alone are insufficient for high-quality prediction, as established across the literature [3][4][15]. Our pipeline will expand these into approximately 60+ features covering technical indicators, lag features, rolling statistics, calendar features, and sentiment-based features. Feature relevance will be formally evaluated using SHAP values in Milestone 5. For the classification task, we expect a roughly balanced 50/50 class distribution across a 10-year period. We will apply SMOTE or class weighting if significant imbalance is detected during EDA in Milestone 3.

### 4.4 Potential Preprocessing Challenges

- Scale normalization: Stock prices vary from ~\$10 to ~\$4,000 across S&P 500 constituents. If a single global scaler were applied across all stocks, high-priced stocks such as Netflix or Booking Holdings would dominate the loss function, causing the model to systematically underfit lower-priced stocks. To address this, Min - Max normalization is applied independently per ticker, so every stock's features are scaled to the same 0 - 1 range regardless of its actual price level. This ensures fair, unbiased learning across all 500 stocks.
- Temporal data leakage: All rolling statistics and lag features must use a strict minimum one-day look-back to prevent future information from entering training features.
- Sentiment alignment: News articles published after 4:00 PM EST will be assigned to the next trading day's sentiment score to prevent future leakage.
- API rate limiting: Bulk downloading 500 tickers will be done in batches with randomized delays, and all data will be cached to disk after first download.

## 5. Preliminary Analytical Framework

This section describes the high-level analytical approach planned for this project, covering the full pipeline from data preprocessing through model training, evaluation, and financial validation.

### 5.1 Proposed Machine Learning Models

**Global Model Architecture (Design Decision):** All five models will be trained as global models - a single model instance trained across all S&P 500 stocks simultaneously, where each (ticker, date) observation is treated as an independent training sample. This design is deliberately chosen over training 500 separate per-stock models for three reasons: (1) a global model can learn cross-sectional patterns and sector-level dynamics that a per-stock model cannot observe; (2) training 500 individual deep learning models is computationally intractable within the project timeline; and (3) per-stock models trained on a single stock's history of approximately 2,500 rows carry a high risk of overfitting, especially for the Transformer and CNN+LSTM architectures. To account for the fact that different stocks behave very differently in terms of price scale and volatility, all features are normalized independently per ticker before pooling into the global training set. A sector identifier is also included as a categorical input feature, allowing the model to learn stock-specific and sector-specific adjustments within the global architecture.

| Model                       | Type                      | Why Selected                                                                                                                                |
|-----------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| LSTM+GRU Hybrid             | Deep Learning (Recurrent) | Benchmark from base article [3]; GRU captures short-term patterns, LSTM refines long - term dependencies                                    |
| Bidirectional LSTM (BiLSTM) | Deep Learning (Recurrent) | Processes sequences in both temporal directions for richer context [5][14]                                                                  |
| CNN+LSTM Hybrid             | Deep Learning (Hybrid)    | CNN extracts local price patterns as shown by Selvin et al. [2]; LSTM handles temporal dependencies                                         |
| Time Series Transformer     | Deep Learning (Attention) | Self-attention enables global temporal context; decoder-only structure confirmed best by Chen [8]; interpretable via attention weights [19] |
| Random Forest               | Ensemble (Tree-based)     | Strong non-linear baseline; interpretable via SHAP [17][18]; validated for stock direction prediction [13]                                  |

### 5.2 Feature Engineering Plan

- Technical Indicators: RSI (14 - day), MACD and Signal Line, Bollinger Bands (upper, lower, width), ATR, OBV, EMA (10/20/50/200 - day), SMA (10/20/50 - day), Stochastic Oscillator (%K, %D), ADX, CCI, Williams %R.
- Lag Features: Closing price lagged by 1, 2, 3, 5, 10, and 20 trading days; lagged volume and lagged 1-day return.
- Rolling Statistical Features: 5/10/20 - day rolling mean, standard deviation, minimum, maximum, and skewness of closing price and volume.
- Calendar and Regime Features: Day of week, month, quarter, earnings season flag, VIX-based market regime indicator.
- Sentiment Features: Daily FinBERT positive, negative, and neutral probability scores; net sentiment score; article count per day.
- Interaction Features: Sentiment  $\times$  5 - day price momentum; sentiment  $\times$  daily trading volume; net sentiment  $\times$  RSI; net sentiment  $\times$  MACD.
- Return-based Features: 1-day, 5-day, and 20-day log returns; realized volatility (rolling 20-day standard deviation of returns).

### 5.3 Preprocessing Techniques

- Per-ticker Min-Max normalization for neural network inputs: Each stock's features are normalized independently to a 0 - 1 range. This is essential because S&P 500 stock prices range from approximately \$10 to over \$4,000 across different companies. Applying a global scaler would cause

high-priced stocks to dominate the training signal. By normalizing per ticker, every stock contributes equally to model learning regardless of its absolute price level.

- Volatility differences across stocks: Beyond price scale, individual stocks differ substantially in daily volatility - for example, high-beta technology stocks may move 3 - 5% per day while defensive utility stocks move less than 0.5%. To ensure the model is not biased toward high-volatility stocks, return-based features (log returns, rolling realized volatility) are used in place of raw prices wherever possible, since percentage returns are naturally comparable across stocks with different volatility profiles.
- Train-Validate-Test split: 70% training (2015 - 2022), 15% validation (2022 - 2023), 15% testing (2023 - 2024).
- Time-series-aware walk-forward cross-validation to prevent data leakage.
- Missing news days filled with neutral sentiment (0.5) and flagged with a binary indicator.

## 5.4 Evaluation Metrics

| Task           | Metric                                   | Target                                      |
|----------------|------------------------------------------|---------------------------------------------|
| Regression     | RMSE (Root Mean Squared Error)           | $\geq 10\%$ improvement over naive baseline |
| Regression     | MAE, MSE, MAPE                           | Reported for all models                     |
| Classification | Directional Accuracy                     | $\geq 60\%$                                 |
| Classification | F1 - Score, Precision, Recall, ROC - AUC | F1-Score $\geq 0.60$                        |
| Financial      | Sharpe Ratio                             | $> 1.0$                                     |
| Financial      | Total P&L Return (%)                     | Positive; beats buy-and-hold                |
| Financial      | Maximum Drawdown, Win Rate               | Reported for all models                     |

## 5.5 Validation Strategy

- Strict chronological split: training data is always earlier than validation and test data.
- Walk-forward backtesting on the held-out test set (2023 - 2024).
- Benchmark comparison against naive price model (prior-day close), random classifier, and buy and hold strategy.
- All five models evaluated under identical conditions to ensure fair comparison.

## 5.6 Visualization and Reporting Tools

- Matplotlib and Seaborn for training curves, feature distributions, and correlation matrices.
- Plotly for interactive equity curves and backtesting performance dashboards.
- SHAP summary plots and dependence plots for Random Forest interpretability.
- Transformer attention weight heatmaps for temporal interpretability.
- Jupyter Notebooks for reproducible, documented experimental results.

## 6. Feasibility and Constraints Assessment

### 6.1 Technical Feasibility

The project is technically feasible within the available timeline and resources. All data is freely accessible via yfinance with no subscription required. All software tools - Python, PyTorch, Scikit-learn, Hugging Face Transformers, and SHAP - are open-source. GPU-accelerated training via Google Colab and Kaggle Notebooks is available at no cost. The main computational challenge of training five architectures on ~1.25 million rows will be managed by first developing and tuning all models on a representative subset of 20 - 30 stocks, then scaling to the full S&P 500 once pipelines are validated.

### 6.2 Hardware and Software Constraints

| Constraint           | Description                                                        | Mitigation Strategy                                                                         |
|----------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| GPU Memory           | Training Transformers and BiLSTM may exceed Colab's 16GB GPU limit | Use gradient accumulation and reduced batch size; use Kaggle's 30 - hour/week GPU as backup |
| API Rate Limits      | Bulk downloading 500 tickers may trigger yfinance rate limiting    | Download in batches with randomized delays; cache all data to disk after first download     |
| Colab Session Limits | Google Colab free tier disconnects after ~12 hours                 | Use Kaggle Notebooks for long runs; save model checkpoints frequently                       |
| FinBERT Inference    | Running FinBERT on thousands of articles per ticker is slow on CPU | Run FinBERT on GPU; cache all sentiment scores to disk after initial computation            |
| Storage              | ~1.25M rows + engineered features may require several GB           | Use compressed CSV and Parquet formats; store on Google Drive for cross-session access      |

### 6.3 Time Management Considerations

| Milestone | Task                                                                  | Planned Completion |
|-----------|-----------------------------------------------------------------------|--------------------|
| 3         | Data collection, yfinance download, sentiment scoring, caching        | June 9, 2026       |
| 4         | Full preprocessing, feature engineering, baseline modelling           | June 23, 2026      |
| 5         | Advanced model training (all 5 architectures), hyperparameter tuning  | July 7, 2026       |
| 6         | Backtesting, financial metrics, SHAP analysis, comparative evaluation | July 21, 2026      |
| 7 & 8     | Oral presentation and final report                                    | August 10, 2026    |

### 6.4 Potential Implementation Challenges and Risk Mitigation

- **Model Overfitting:** Dropout regularization, early stopping, and L2 weight decay will be applied to all deep learning models. Walk-forward cross-validation will detect and prevent data leakage throughout training.
- **Temporal Data Leakage:** All rolling, lag, and sentiment features will use a strict minimum one-day lag. The 60/20/20 chronological train-validate-test split will be enforced throughout the project.
- **Scope Creep:** A feature and model freeze will be enforced after Week 4. Ideas beyond defined scope will be logged as future work and not implemented during this project.
- **Team Coordination:** A shared GitHub repository and task board will be maintained to ensure parallel workstreams and consistent progress tracking.

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